SVD

What you need to know -

1. $AV = U\Sigma$
2. $A = U\Sigma V^T$
3. $AA^T = U\Sigma\Sigma^T U$
4. $A^T A = V\Sigma^T\Sigma V^T$

Example 1 :

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \qquad A^T = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix}$$

$$A.A^T = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

$$\lambda^2 - 34\lambda + 225 = 0$$

$$A^T.A = \begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix}$$

$$\lambda^3 - 34\lambda^2 + 225\lambda - 0 = 0$$

$$\boxed{\lambda_1 = 25,\ \lambda_2 = 9,\ \lambda_3 = 0}$$

For $AA^T \rightarrow U$,

$$0 = (A.A^T - 25I)x = \begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad u_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$0 = (A.A^T - 9I)x = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad u_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

For $A^TA \rightarrow V$,

$$0 = (A^T.A - 25I)x = \begin{bmatrix} -12 & 12 & 2 \\ 12 & -12 & -2 \\ 2 & -2 & -17 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \qquad v_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$0 = (A^T.A - 9I)x = \begin{bmatrix} 4 & 12 & 2 \\ 12 & 4 & -2 \\ 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} \qquad v_2 = \begin{bmatrix} 1/3\sqrt{2} \\ -1/3\sqrt{2} \\ 4/3\sqrt{2} \end{bmatrix}$$

$$0 = (A^T.A - 0I)x = \begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1/2 \end{bmatrix} \qquad v_2 = \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix}$$

$$U = \begin{bmatrix} \uparrow & \uparrow \\ u_1 & u_2 \\ \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$V = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ v_1 & v_2 & v_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/3\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -1/3\sqrt{2} & -2/3 \\ 0 & 4/3\sqrt{2} & 1/3 \end{bmatrix}$$

$$\lambda = \delta^2 \Rightarrow \delta = \sqrt{\lambda}$$

$$\therefore \delta_1 = \sqrt{25} = 5 \qquad \delta_2 = \sqrt{9} = 3$$

$$A = \underset{(2\times 2)}{\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}} \underset{\substack{(2\times 3) \\ \text{(To match dimension)}}}{\begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}} \underset{(3\times 3)}{\begin{bmatrix} 1/\sqrt{2} & 1/3\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -1/3\sqrt{2} & -2/3 \\ 0 & 4/3\sqrt{2} & 1/3 \end{bmatrix}^T}$$

$$\boxed{A = U \Sigma V^T}$$

Do not mess up the orders

Verify